

Dongfang Zhao, Ph.D.

Nationality: United States of America

Education

University of Washington (UW), Seattle	Computer Science, PostDoc	2017
Illinois Institute of Technology, Chicago	Computer Science, PhD	2015
Emory University, Atlanta, GA	Computer Science, MS	2008

Professional Experience

2026–now	Open-source Contributor, Meta/Facebook Faiss (CUDA main branch)
2025–now	Open-source Contributor, Microsoft DiskANN (C++ main branch)
2025–now	Adjunct Assistant Professor, UW Paul G. Allen School of Computer Science & Engineering
2023–now	Tenure-track Assistant Professor, UW Tacoma School of Engineering & Technology
2023–2025	Visiting Faculty, UW Paul G. Allen School of Computer Science & Engineering
2018–2023	Affiliate Faculty, University of California, Davis
2018–2023	Assistant Professor, University of Nevada, Reno (UNR)
2015–2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014–2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013–2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer (full-time), Epic Systems Corporation, Madison, WI

Selected Awards

2022	Federal Government Grant, UNR PI, NSF 2112345, \$255,916 “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Government Grant, UNR PI, DOE SC0020455, \$200,000 “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000
2007	Graduate Fellowship, Oak Ridge Institute for Science and Education, \$85,000

Recent Projects

[ElasticIVF]	Elastic in-memory vector indexing and search for streaming data analysis on heterogeneous distributed platforms: https://github.com/hpdic/ElasticIVF <i>Tags:</i> Distributed GPU computing, Message passing interface (MPI), Faiss, CUDA kernel
[AdaDisk]	Decentralized agentic systems for adaptive ingestion-query scheduling of disk-resident vector indices in LLM serving: https://github.com/hpdic/AdaDisk <i>Tags:</i> Large language models, AI agents, Distributed systems, DiskANN, C++, Python
[Hermes]	MySQL plug-in for high-performance homomorphic encryption of vector databases in privacy-preserving RAG: https://github.com/hpdic/Hermes <i>Tags:</i> Retrieval-augmented generation, Fully homomorphic encryption, OpenFHE, C++

Recent Publications (Dongfang Zhao was/is the *student's academic advisor)

- [1] *Jiuzhou Fu, **Dongfang Zhao**. QPAD: Quantile-Preserving Approximate Dimension Reduction for Nearest Neighbors Preservation in High-Dimensional Vector Search. 42nd IEEE International Conference on Data Engineering (ICDE), 2026.
- [2] *Jiayao Wang, ..., **Dongfang Zhao**. BadRSSD: Backdoor Attacks on Regularized Self-Supervised Diffusion Models. The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- [3] *Baigang Chen, **Dongfang Zhao**. Reliable Non-Leveled Homomorphic Encryption for Web Services. The ACM Web Conference (formerly International World Wide Web Conference, or WWW), 2026.
- [4] *Chengyu Gong, *Gefei Shen, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. Order-Preserving Dimension Reduction for Multimodal Semantic Embedding. The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026.
- [5] *Jiuzhou Fu, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. ProHD: Projection-Based Hausdorff Distance Approximation. IEEE International Conference on Data Mining (ICDM), 2025.
- [6] *Jiayao Wang, ..., **Dongfang Zhao**. IPBA: Imperceptible Perturbation Backdoor Attack in Federated Self-Supervised Learning. European Conference on Artificial Intelligence (ECAI), 2025.
- [7] *Abdullah Al-Mamun, **Dongfang Zhao**, et al. ZTP: A Scalable and Lightweight Privacy-Preserving Blockchain via Scale-Free Quorums and Geometric Fragmentation. International Conference on Parallel Processing (ICPP), 2025.
- [8] *Gefei Shen, **Dongfang Zhao**. Towards Nonparametric Topological Layers in Neural Networks. The 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024.
- [9] *Muhammad Jahanzeb Khan, ..., **Dongfang Zhao**. Privacy-Preserving Artificial Intelligence on Edge Devices: A Homomorphic Encryption Approach. IEEE International Conference on Web Services (ICWS) 2024: 395-405.
- [10] *Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data (SIGMOD), 2023.
- [11] *Abdullah Al-Mamun, **Dongfang Zhao**. ECHAIN: Securing Electronic Device Provenance through Privacy-Preserving Consortium Blockchains. The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023.
- [12] *Olamide Timothy Tawose, *Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [13] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [14] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021.
- [15] *Xinying Wang, *Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020.